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nominal money supply is exogenous and equal to M_0 ; thus the real money supply is (M_0/P) . In this problem C, Y, r and P denote consumption, real GDP, interest rate and price level respectively.

(i) Set up the IS-LM equations.

(ii) Determine how an increase in the price level would affect real GDP and the interest rate?

9. Let x and y be two random variables having $\text{Var}(x) = 4$ and $\text{Var}(y) = 9$. If $\text{Cov}(x, y) = 2$, then is $\text{Var}(x - y)$ larger or smaller than it is when x and y are independent? What is your answer if $\text{Cov}(x, y) = -2$?
10. Of six candidates seeking 3 positions in a counseling center, 2 have degrees in social sciences and 4 do not. If 3 candidates are selected at random, find the probability distribution of X , where X is the random variable defined as the number of candidates having social sciences degree among the selected persons.

R2 Subhorup Chattopadhyay

Test Code: ME I/ME II 2002

Syllabus for ME I

2002

Matrix Algebra: Matrices and Vectors, Matrix Operations, Determinants, Non-singularity, Inversion, Cramer's rule.

Calculus: Limits, Continuity, Differentiation of functions of one or more variables, Product rule, Partial and total derivatives, Derivatives of implicit functions, Unconstrained optimization (first and second order conditions for extrema of functions of single variable and several variables), Taylor Series, Definite and Indefinite Integrals: standard formulae, integration by parts and integration by substitution. Differential equations, Constrained optimization of functions of a single variable.

Theory of sequence and series.

Linear programming: Formulations, statements of Primal and Dual problems, Graphical solutions.

Theory of polynomial equations (up to third degree)

Sample Questions for ME I (Mathematics) 2002

1. The function $f(x) = |x|$ is

- (A) both continuous and differentiable at $x=0$
 (B) neither continuous nor differentiable at $x=0$
 (C) continuous, but not differentiable at $x=0$
 (D) differentiable, but not continuous at $x=0$

2. Let $f(x)$ be a function satisfying the condition $f(x) = f(-x)$ for all real x .

If $f'(0)$ exists, then $f'(0)$ will be

- (A) 0 (B) -1 (C) 1 (D) none of these

3. Let $a_n = \frac{1}{n}$ and $b_n = \frac{1}{n^2}$ be two given sequences. Then, $\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$ is

- (A) = 0
 (B) = $+\infty$
 (C) not defined
 (D) = 1

4. Suppose a time-dependent variable, $X(t)$, changes over time according as $dX(t)/dt = 3t^2$. Then, at any time t , $X(t)$ depends on its initial value $X(0)$ in the following way:

- (A) $X(0) + 3t^{3/2}$ (B) $X(0) + 3t^{3/2}$
 (C) $X(0) + 2t^{3/2}$ (D) $X(0) + 2t^{3/2}$

5. The second-order Taylor expansion of the function $\frac{1}{1+x}$ around

$x_0 = 1$ is:

- (A) $\frac{1}{2} + \frac{1}{2(1+x)} + \text{remainder}$
 (B) $\frac{1}{2} - \frac{1}{8}(x-1) + \frac{1}{4}(x-1)^2 + \text{remainder}$
 (C) $\frac{1}{2} + \frac{1}{2}(x-1) + \frac{1}{8}(x-1)^2 + \text{remainder}$
 (D) $\frac{1}{2} - \frac{1}{4}(x-1) + \frac{1}{8}(x-1)^2 + \text{remainder}$

6. The limit of the sequence $+1, -1, +1, -1, \dots$

- (A) does not exist
 (B) is 1
 (C) is -1
 (D) is 0

7. Let A and B be $n \times n$ matrices such that the determinant of AB is zero. Then which of the following statements must be true?

- (A) determinant of $A+B$ is zero
 (B) A and B are both nonsingular

(C) At least one of A and B is singular

(D) $AB = BA$

8. The matrix $\begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$ is nonsingular only if

(A) $a \neq b, b \neq c, c \neq a$

(B) $a \neq b$ or $b \neq c$ or $c \neq a$

(C) $a+b+c \neq 0$

(E) none of these

9. The solution to the problem, $\max_{x,y} (2x^2 + y^2)$ subject to

$x + y \leq 1, x, y \geq 0$ is given by

(A) $x = \frac{1}{2}$

(B) $x = \frac{2}{3}$

(C) $x = 1$

(D) none of the above

10. In what ratio should a given line be divided into two parts so that the area of the rectangle contained by them is maximum?

(A) 1:1 (B) 4:3 (C) 3:2 (D) $(\sqrt{5}+1) : (\sqrt{5}-1)$

11. The derivative of $\int_a^{100} x dx$ with respect to a is:

(A) a

(B) $-a$

(C) a^2

(D) $100a$

12. Consider the sequence 1, 3, 8, 19, 42, ...; the number next to 42 is:

(A) 65

(B) 75

(C) 89

(D) 101

13. The equation of the tangent to the graph of $f(x) = x^2$ at $x = 3$ is:

(A) $y = 6x - 9$

(B) $y = 3x$

(C) $y = 12x - 27$

(D) none of the above.

14. What is the sum of the series $1^2 + 2^2x + 3^2x^2 + 4^2x^3 + \dots$, where

$0 < x < 1$?

(A) 1

(B) $(1-x)^{-1}$

(C) $(1+2x)(1+x)^{-1}$

(D) $(1+x)(1-x^3)^{-1}$

15. Consider the following linear programming problem:

Maximize $x_1 + x_2$

Subject to $x_1 + 2x_2 \leq 4$

$x_1 + 6x_2 \leq 6$

$x_1 - 2x_2 \leq 2$

$x_1, x_2 \geq 0$

An optimal solution to the problem is

(A) $x_1 = 4, x_2 = 0$

(B) $x_1 = 0, x_2 = 1$

(C) $x_1 = 3, x_2 = \frac{1}{2}$

(D) none of these

Syllabus for ME II

Microeconomics: Consumer theory, Producer theory, Market Forms, and Welfare Economics

Macroeconomics: National income accounting, Simple model of income determination and multiplier, IS-LM model, Banking, money supply and inflation.

Elementary Statistics: Measures of central tendency, dispersion, correlation, Elementary probability theory.

Sample Questions for ME II (Economics) 2002

1. Instruction for question numbers 1 (i) – 1 (iv)

For each of the following questions four alternative answers are provided. Choose the answer that you consider to be the most appropriate for a question and write it down in your answer book.

(i) Suppose that a consumer consumes only two goods, x_1 and x_2 . If

the income elasticity of x_1 is 1.2, then the income elasticity of x_2 is

(A) greater than 1.2;

(B) less than 1.2 but greater than 1; ✓

✓ (C) less than 1;

(D) exactly 1.2.

- (ii) If the government imposes a 10% tax on a monopolist's profit, then the monopolist will

(A) decrease his output by greater than 10%;
 (B) decrease his output by 10%;
 (C) not change his output;
 (D) decrease his output by less than 10%.

- (iii) Which of the following transactions is not included in GDP?

(A) Mr. Ramgopal, a private investment banker, hires Mr. Gopi to do cooking and cleaning at home.
 (B) Mr. Ramgopal buys a new Maruti Esteem.
 (C) Mr. Ramgopal flies from Delhi to Kolkata to see Durga Puja
 (D) Mr. Ramgopal directly buys 100 stocks of Satyam Enterprises.

- (iv) Suppose the population of an economy is divided into two groups, A and B. A proportion α of aggregate income accrues to the former and the rest to the latter. Suppose the former have a higher consumption propensity as well as a lower propensity to demand money with respect to income. What happens to real GDP (Y) and interest rate (r), in terms of the IS-LM model, if the distribution of income shifts from A to B?

(A) Y increases r falls;
 (B) Y decreases and r rises;
 (C) Y may increase or fall while r falls;
 (D) Y decreases while r may rise or fall.

- 2.(a) Suppose a consumer always consumes 2 spoons of sugar with each cup of tea. If the price of sugar is Rs. p_1 per spoon, the price of tea is Rs. p_2 per cup and the consumer has Rs. y to spend on tea and sugar, determine how much of tea and sugar she will want to consume?

- (b) Show that exactly two of the following utility functions give rise to the same demand functions for goods 1 and 2

(i) $u(x_1, x_2) = \sqrt{x_1 x_2}$.

(ii) $u(x_1, x_2) = x_1^2 x_2$.

(iii) $u(x_1, x_2) = \ln x_1 + \ln x_2$.

- 3.(a) Consider a profit-maximizing firm in the short run. It uses a variable factor x_1 and a fixed factor x_2 . If the price of x_2 goes down, what happens to the x_1 quantity demanded of x_1 ? What happens to the Firm's level of profit?

$2x = y$
 $p_1 x + p_2 x = y$
 $x = y / (2p_1 + p_2)$
 $E = y / (2p_1 + p_2)$

- (b) The marginal cost rises by 5% and the monopolist raises the price by 5%. Give a condition under which this is consistent with profit maximizing behaviour.

4. The total cost function of a competitive firm, which sells two products are given by $C(q_1, q_2) = 2q_1^2 + q_1q_2 + 2q_2^2$. The prices are Rs.12 and Rs.18 respectively for good 1 and good 2. What are the profit maximizing levels of output?

5. Consider a society where people throw garbage on the streets. A person who does so inflicts negative externalities on others. Suppose the rupee cost of this externality to any one person is $a \times n$, where a is a positive constant and n is the number of (other) people who throw garbage on the streets. Assume that your own act of throwing garbage on the street gives you a gain worth G rupees and the rupee-loss due to shame is equal to (C/n) where C is a positive constant. Show that

- (a) if no one else throws garbage on the streets, it is optimal for you not to throw garbage on the streets;
(b) if everyone else throws garbage on the streets, then there is a population size n^* , such that if n , the number of other people

throwing garbage on the streets, is greater than or equal to n^* , it is optimal for you to throw garbage on the streets. What does n^* depend on?

6. Consider the IS function, wherein consumption depends on income only, investment depends on interest only and government spending is exogenous. Suppose that, as a consequence of a unit increase in government expenditure, the IS curve is found to shift vertically by 0.5 at each value of GDP. What is the rate of change of investment with respect to the rate of interest?
7. Consider a simple Keynesian (closed) economy as described below. Aggregate supply function is given by $Y = f(N)$, relating output to employment. The consumption function is $C = aY$ and $f' > 0 > f''$ and $0 < a < 1$. Aggregate expenditure is $E = C + \bar{I}$ and the equilibrium condition is: $Y = E$. Find the "employment multiplier" $(dN/d\bar{I})$.
8. Consider the IS-LM model. Let the consumption function be given by $C = a + bY$, $a > 0$, $0 < b < 1$. Investment and Government spending are exogenous and given by I_0 and G_0 respectively. In the money market, the real demand for money is given by $L = ky - gr$, $k > 0$, $g > 0$. The